SECTION A

1. (a) Consider the following subset of the real numbers

$$S = (-\infty, 5) \cup \{2020\}$$

(i) Is the set S open? Is the set S closed? In each case justify your answer.

(ii) Determine the set of limit points of S and the set of interior points of S. (10)

Solution: Seen many similar exercises like this, in class and in tutorials

(i) The set S is not open because $x_0 = 2020$ is not an interior point of S. It is not closed because it does not contain all its limit points, for example it does not contain 5 which is a limit point.

(ii) The set of limit points of S is given by $(-\infty, 5]$. The set of interior points is given by $(-\infty, 5)$.

(b) Prove, using the ϵ definition of limit, that

$$\lim_{x\to\infty} \frac{1}{x^3 + 2x + 1} = 0.$$

(10)

Solution: Seen many similar exercises like this, in class and in tutorials

Fix $\epsilon > 0$. Then for $\beta = \sqrt[3]{\frac{1}{\epsilon}}$ we get that if $x > \beta$, then $\left|\frac{1}{x^3+2x+1}\right| \leq \left|\frac{1}{x^3}\right| < \frac{1}{\beta^3} = \epsilon$, as we wanted to show.

(c) Suppose that $\{x_i\}_{i=0}^{\infty}$ and $\{y_i\}_{i=0}^{\infty}$ with $x_i, y_i \in \mathbb{R}^n$ are two sequences of vectors with

$$x_i \to x, \quad y_i \to y \quad \text{as } i \to \infty.$$

Show that

$$x_i + y_i \to x + y \quad \text{as } i \to \infty.$$

(10)

Solution: (Seen similar (for converging functions in R^n)) Fix $\epsilon > 0$. By the convergence of $\{x_i\}$ and $\{y_i\}$ there exists N_1 and N_2 such that for $i \geq N_1$ and $j \geq N_2$ we have

$$\|x_i - x\| < \frac{\epsilon}{2}$$

and

$$\|y_i - y\| < \frac{\epsilon}{2}.$$

Set $N = \max\{N_1, N_2\}$. Then if $i \geq N$ we have

$$\|x_i + y_i - (x + y)\| \leq \|x_i - x\| + \|y_i - y\| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

(d) Give, with proof, the value(s) of $p \in \mathbb{R}$ for which the following improper integral converges

$$I = \int_0^1 \frac{1}{x^p} \mathrm{d}x.$$

Solution: (unseen) For $p \neq 1$, we compute, using the Fundamental Theorem of Calculus,

$$\int_\epsilon^1 \frac{1}{x^p} \mathrm{d}x = \left[\frac{x^{1-p}}{1-p}\right]_\epsilon^1 = \frac{1}{1-p}(1 - \epsilon^{1-p}).$$

If $p < 1$, then $\epsilon^{1-p} \to 0$ as $\epsilon \to 0$ and the integral converges. If $p > 1$, then $\epsilon^{1-p} \to \infty$ as $\epsilon \to 0$ and the integral does not converge. It remains to check the case $p = 1$. Here,

$$\int_\epsilon^1 \frac{1}{x} \mathrm{d}x = [\log x]_\epsilon^1 = -\log \epsilon \to \infty$$

as $\epsilon \to 0$, so again the integral does not converge. Therefore, the integral converges if and only if $p < 1$. (10)

(e) (i) State the Inverse Function Theorem for a function $g : \mathbb{R}^n \to \mathbb{R}^n$.

(ii) Let $h : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$h(x, y) = \begin{pmatrix} x + 5y^5 \\ 7xy - y \end{pmatrix}.$$

Using the Inverse Function Theorem, show that there is a neighbourhood u of $(0, 0)$ such that $\forall q \in u$, the equation $h(p) = q$ is soluble. (10)

Solutions:

(i) (Theorem central to course) The inverse function theorem states that if $g : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable in an open set containing p and the differential $Dg(p)$ is

invertible, then there exist open sets $U, V \subset \mathbb{R}^n$ with $p \in U$ and $g(p) \in V$ such that $g : U \to V$ is bijective. Moreover, the inverse function $g^{-1} : V \to U$ is continuously differentiable and $Dg^{-1}(g(p)) = (Dg(p))^{-1}$.

(ii) (unseen) First note that $h(0) = 0$. Secondly, note that all of the partial derivatives $D_i h^j(p)$ for $i, j = 1, 2$ exist and are continuous in p, so that h is continuously differentiable everywhere in $\mathbb{R}^2$ by a result from the course. Now, setting $p = (x, y)^t$, we have (making the usual identification of a linear map with its matrix with respect to the canonical basis)

$$Dh(p) = \begin{pmatrix} 1 & 25y^4 \\ 7y & 7x - 1 \end{pmatrix}$$

so that

$$Dh(0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This is clearly invertible, so we apply the inverse function theorem to deduce that there exist open sets $U, V \subset \mathbb{R}^2$ with $0 \in U$ and $0 \in V$ such that $h : U \to V$ is a bijection. Since h maps U bijectively onto V there exists $p \in U$ with $h(p) = q$ with $q \in V$. **[50]**

SECTION B

2. We consider the function

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

where $a_i \in \mathbb{R}$ for all $i = 0, \ldots, n-1$.

(a) Prove that $p(x)$ is bounded on $\mathbb{R}$ if and only if $n = 0$. (6)

(b) If n is odd prove that for all $\alpha \in \mathbb{R}$ there exists $x_0 \in \mathbb{R}$ such that $p(x_0) = \alpha$. (8)

(c) If n is even prove that there exist x_0 and x_1 such that $x_0 \neq x_1$ and such that $p(x_0) = p(x_1)$. (11)

[25]

Solution: The first part is unseen, but really easy. As for the second question, in the tutorials they have seen the case of a polynomial of degree 3. The third part is challenging.

(a) If $n = 0$, then $p(x) = a_0$ which is bounded. Viceversa suppose that $n \neq 0$. Then $\lim_{x\to\infty} p(x) = \infty$, so $p(x)$ is not bounded.

(b) Since $\lim_{x\to\infty} p(x) = \infty$ and $\lim_{x\to-\infty} p(x) = -\infty$, and p is continuous on $\mathbb{R}$ then there exists $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) < \alpha$ and $f(x_2) > \alpha$. By the intermediate value theorem there exists x_0 between x_1 and x_2 such that $f(x_0) = \alpha$.

(c) First we prove that $p(x)$ is bounded from below. Since n is even $\lim_{x\to\infty} p(x) = \infty$ and $\lim_{x\to-\infty} p(x) = \infty$ which means that for all $M > 0$ there exists $\delta > 0$ such that if $|x| > \delta$, then $p(x) > M$. We consider now the function $p(x)$ on $[-\delta, \delta]$, which is closed and bounded, so $p(x)$ attains minimum and maximum on it. Let x_0 the point in which $p(x)$ attains its minimum and denote $N = p(x_0)$. We define a function $q(x)$ in the following way

$$q(x) = \begin{cases} p(x) - 2N & \text{if } N > 0 \\ p(x) & \text{if } N < 0 \\ p(x) & \text{if } N = 0 \end{cases}$$

We remark that $q(x_0) < 0$ and $\lim_{x\to\infty} q(x) = \infty$ and $\lim_{x\to-\infty} q(x) = \infty$. Hence by the intermediate value theorem on $[-a, x_0]$ there exists x_1 such that $q(x_1) = 0$ and on $[x_0, a]$ there exists x_2 such that $q(x_2) = 0$. Therefore $x_1 \neq x_2$ and $p(x_1) = p(x_2)$.

3. (a) State the Riemann criterion for integrability. (5)

(b) Let $f : [a, b] \to \mathbb{R}$ be an increasing function. Define the upper and lower Riemann sums of f, and use them to show that f is Riemann-integrable on $[a, b]$. (8)

(c) Compute the limit

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \sin\left(\frac{k\pi}{2n}\right)}{n}.$$

(12)

Solution: [**25**]

(a) (seen theorem central to course) A function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if $\forall\, \varepsilon > 0$, $\exists$ a partition P such that

$$U(f; P) - L(f; P) < \varepsilon.$$

where $U(f; P)$ and $L(f; P)$ denote the upper and lower Riemann sums respectively.

(b) (seen in exercise sheet) Given $n \in \mathbb{N}$, let $P = \left(a + \frac{i(b-a)}{n}\right)_{i=0,\ldots,n}$. Then

$$L(f; P) = \sum_{i=0}^{n-1} \frac{b-a}{n} f\left(a + \frac{i(b-a)}{n}\right)$$

and

$$U(f; P) = \sum_{i=1}^{n} \frac{b-a}{n} f\left(a + \frac{i(b-a)}{n}\right).$$

So then, $U(f; P) - L(f; P) = \frac{b-a}{n}(f(b) - f(a))$. This tends to 0 as $n \to \infty$, so using the Riemann criterion for integrability, f is Riemann integrable.

(c) (seen similar) Observe that

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \sin\left(\frac{k\pi}{2n}\right)}{n} = \frac{2}{\pi} \sum_{k=1}^{n} \sin\left(\frac{k\pi}{2n}\right) \frac{\pi}{2n} = \frac{2}{\pi} U(f; P)$$

where $U(f; P)$ is the upper Riemann sum for the function f and P is the partition of $[0, \frac{\pi}{2}]$ into n subintervals of equal length. Since f is interchangeable it follows that

$$\lim_{n\to\infty} U(f; P) = \int_0^1 \sin x \mathrm{d}x.$$

Then by the Fundamental Theorem of Calculus and the fact that $(-\cos x)' = \sin x$ it follows that

$$\int_0^1 \sin x \mathrm{d}x = \cos(0) - \cos(\pi/2) = 1.$$

Hence

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \sin\left(\frac{k\pi}{2n}\right)}{n} = \lim_{n\to\infty} \frac{2}{\pi} U(f;P) = \int_0^1 \sin x \mathrm{d}x = 1.$$

4. Let λ be a real number such that $\lambda \in \left(-\frac{e}{2}, \frac{e}{2}\right)$. We consider the function

$$f(x) = e^x - \lambda x^2.$$

(a) Let $g(x) = f'(x)$. Prove that $g(x)$ is Lipschitz continuous on $[0,1]$ and compute the smallest Lipschitz constant L for $g(x)$ on $[0,1]$. (5)

(b) Compute

$$\lim_{x\to\infty} f(x)$$

and

$$\lim_{x\to-\infty} f(x)$$

(7)

(c) How many solutions has the equation

$$f(x) = 0?$$

Justify your answer. (13)

[**25**]

Solution: Part (i) is similar to many examples seen in tutorials, part (ii) is an easy calculation. Part (iii) is challenging.

(a) We compute $f'(x) = g(x) = e^x - 2\lambda x$. Since $g \in \mathcal{C}^1([0,1])$, it is Lipschitz continuous on $[0,1]$ and the smallest Lipschitz constant is $L = \sup_{x\in[0,1]} g'(x) = \sup_{x\in[0,1]} e^x - 2\lambda = e - 2\lambda$ because $g'(x)$ is strictly increasing in $[0,1]$ ($g''(x) = e^x > 0$ for $x \in [0,1]$).

(b) If $\lambda = 0$, then $\lim_{x\to\infty} e^x - \lambda x^2 = \infty$ and $\lim_{x\to-\infty} e^x - \lambda x^2 = 0$. If $-\frac{e}{2} < \lambda < 0$, then $\lim_{x\to\infty} e^x - \lambda x^2 = \infty$ and $\lim_{x\to-\infty} e^x - \lambda x^2 = \infty$. If $0 < \lambda < \frac{e}{2}$, then $\lim_{x\to\infty} e^x - \lambda x^2 = \infty$ and $\lim_{x\to-\infty} e^x - \lambda x^2 = -\infty$.

(c) If $\lambda = 0$ the equation has no solution.

We consider now the case of $0 < \lambda < \frac{e}{2}$. In this case $\lim_{x\to\infty} e^x - \lambda x^2 = \infty$ and $\lim_{x\to-\infty} e^x - \lambda x^2 = -\infty$. We want to study the derivative $f'(x) = e^x - 2\lambda x$. If $0 < \lambda < \frac{e}{2}$, then $\lim_{x\to\infty} f'(x) = \infty$ and $\lim_{x\to-\infty} f'(x) = \infty$. We derive again $f''(x) = e^x - 2\lambda$, and $f''(x) = 0$ if and only if $x = \log 2\lambda$; and $f'(\log(2\lambda)) = 2\lambda - 2\lambda \log 2\lambda = 2\lambda(1 - \log 2\lambda)$, so $f'(\log(2\lambda)) > 0$ because we are in the case of $0 < \lambda < \frac{e}{2}$. We can then conclude that $f'(x) > 0$ for all $x \in \mathbb{R}$. Since $\lim_{x\to\infty} e^x - \lambda x^2 = \infty$ and

$\lim_{x\to-\infty} e^x - \lambda x^2 = -\infty$ and $f'(x) > 0$ for all $x \in \mathbb{R}$ we can conclude the equation

$$e^x - \lambda x^2 = 0$$

has one solution.

We consider now the case of $\lambda < 0$. Let us denote $f(x) = e^x - \lambda x^2$. In this case $f(x) > 0$ for all $x \in \mathbb{R}$, so the above equation has no solutions.